OUR PROPOSAL:

Introduction

Power-constrained applications that utilize vision, such as drones, have various sensor options to choose from. Frame-based CMOS image sensors (CIS) capture complete images at a fixed rate, paying a roughly constant power cost regardless of scene content. Event-based dynamic vision sensors (DVS) only report per-pixel brightness changes, so power scales with scene activity. This paper will look into the natural comparison question, *for a given task and scene, which sensor is more power-efficient?*

In this document, we will look at two models. ModuCIS provides a modular, device-to-system power/latency/noise model for frame-based CIS. Additionally, a DVS model captures static power plus dynamic power proportional to event rate. However, no unified framework exists to compare them under realistic, task-driven constraints (Limitation 3 in the project document). Furthermore, neither model currently translates task requirements into sensor configurations (Limitation 1) or predicts event rates from scene dynamics (Limitation 2).

Our project addresses all three limitations using <u>object tracking as the target task</u>. We will build a scene model, drive both sensor models from it, and produce quantitative design rules (for example "DVS is preferred when object velocity exceeds X and background complexity stays below Y; CIS is preferred otherwise").

Aspects to Extend

There are three key limitations that our team will be addressing

<u>Gap 1: Weak Algorithm-to-Sensor-Coupling</u>

The CIS model can sweep hardware parameters, but does not automatically translate tracking requirements into sensor settings. We address this by deriving min. CIS configurations directly from tracking constraints:

- Minimum FPS to track an object
- Minimum SNR to detect an object against a background
- Minimum Resolution set by the smallest detectable object size

<u>Gap2: DVS Model Not Task Aware</u>

The DVS power model takes event rate as an input, but does not predict event rate from the scene. Thus, we can derive the event rate analytically from scene parameters:

- Event Rate
    - For slow object: low event rate → low DVS power
    - For fast object: high event rate → high DVS power
- False positive events from background noise (wasted energy) and false negatives from high thresholds (missed information)
- Enforce refractory-period throughput cap: if scene is too busy, DVS saturates

Gap3: No Unified Comparison Protocol

There are no evaluation setups for comparison between CIS and DVS. We will follow fixed baseline values for both the sensors

- Same pixel area
- Same scene
- Same success metrics

Proposed approaches and methods

Scene Model (shared input to both sensors)

We construct a parametric scene model consisting of:

- Moving an object of a pre-defined size
- Object velocity
- Two types of backgrounds
- Different motion directions

ModuCIS

- Use tracking constraints from the scene model as inputs to ModuCIS.
- Translate minimum frame rate, resolution, and SNR into CIS sensor configurations.
- Sweep key CIS design knobs: frame rate, image resolution, and ADC bit depth.
- Extract total power, latency, and noise/SNR for each configuration.
- Identify minimum-power CIS operating points that satisfy tracking requirements.
- Provide CIS power–performance trends for unified CIS–DVS comparison and final design rules.

DVS

- Use tracking scene parameters as inputs to the DVS power model.
- Translate object velocity, edge density, and motion direction into predicted event rate.
- DVS design knobs: event threshold, filtering policy, and false event assumptions (FP/FN).
- Model effective event rate under refractory-limited throughput and potential saturation.
- Compute total DVS power using static power plus dynamic energy per event.

Comparison + Design Rules

- plot CIS power vs DVS power as a function of object velocity (identifying the crossover speed where one becomes cheaper than the other)
- Examine effect of background/object sensitivity (does crossover point shift when background goes from clean to cluttered)
- how does the crossover change when CIS resolution changes, or DVS threshold change
- at what scene activity level does DVS power exceed CIS power

- A ‘design rule’ in the form of "For this tracking task, DVS is preferred when velocity > V and background complexity < D_max; CIS is preferred when….”

Plan out which tasks will be involved

Task (T)	Owner	Dependencies
T1: Scene Model	Ramaa	None
T2: ModuCIS	Harshitha	T1
T3: DVS	Ish	T1
T4: Comparison & design rule	Sergey	T2+T3
T5: Write-up	ALL	T4

Work distribution among the group
Sergey: CIS/DVS comparison, crossover visualization, design rule, overall project coordination
Harshitha: ModuCIS., parameter sweeps, CIS power/latency/SNR analysis
Ish: DVS, event-rate derivation, DVS power analysis
Ramaa: Scene model construction, analytical formulas linking scene to sensor requirements

<u>TA’s FEEDBACK:</u>

This is a strong proposal that correctly identifies key gaps from the course baseline. The choice of object tracking as the target task is appropriate given its meaningful dynamics for CIS vs DVS comparison. Task decomposition is clear, and the design rule output format aligns well with course expectations.

A few suggestions to strengthen the proposal:

1. Provide more specificity on scene model parameters like include concrete values for object sizes, velocities, motion patterns, and background types rather than vague descriptions.
2. Include analytical formulas for deriving sensor requirements from tracking constraints.
3. Starting from your proposed synthetic approach (systematically varying object size, velocity, motion, and background), consider developing more sophisticated strategies to mimic realistic cases. For example, you could convert or bridge existing real-world object tracking datasets to validate your design rules.

Task-Driven Hardware Optimization of CIS and DVS Image Sensor Architectures

Team members: Harshitha Umesh, Ishita Malik, Ramaa Nitin Vaidya, Sergey Petrushkevich

Introduction

Modern vision systems rely on algorithm–hardware co-design rather than pixel-level improvements alone. Sensor parameters such as frame rate, resolution, ADC precision, and event thresholds strongly affect power, latency, and task performance.
This project builds on ModuCIS (frame-based CMOS image sensor modeling) and a DVS power model. While these tools support basic parameter sweeps, they do not provide a task-driven method to identify optimal hardware configurations. Our project extends this baseline by systematically finding the best CIS and DVS settings for a chosen vision task under performance and power-efficiency objectives.

Aspect to Be Extended

The background material highlights limited algorithm-to-sensor coupling and lack of automatic optimization. We extend this work by introducing a task-driven design-space exploration framework that determines optimal CIS and DVS configurations under two goals:

1. **Best task performance (allowing higher power)**
2. **Best power efficiency while maintaining acceptable task quality**

Proposed Approach and Methods

Selected Vision Task
[Insert chosen task, e.g., motion detection]
A fair comparison setup will be used with identical assumptions (supply voltage, latency budget, and task conditions).
CIS Optimization (ModuCIS)
We will sweep:

- **Frame rate**
- **Resolution**
- **ADC resolution**

For each configuration, we will record power, latency, and task-quality proxy.

DVS Optimization

We will sweep:

- **Event threshold**
- **Filtering policy**
- **False event assumptions**

For each configuration, we will measure event rate, static/dynamic power, and quality proxy.

For both CIS and DVS, we will identify:

- **Performance-first configuration**
- **Power-efficient configuration**

Results will be summarized using Power vs Performance plots and tables.

Tasks Involved

1. **Study baseline CIS and DVS models**
2. **Reproduce reference outputs**
3. **Define task metrics**
4. **Run CIS parameter sweeps**
5. **Run DVS parameter sweeps**
6. **Analyze power–performance tradeoffs**
7. **Generate plots and tables**
8. **Prepare report and presentation**

Work Distribution

Member	Responsibility
Member 1	Task definition and baseline setup
Member 2	CIS parameter sweeps
Member 3	DVS modeling and sweeps
Member 4	Visualization, optimization summary, report

All members will contribute to final analysis and conclusions.

Expected Outcome

The project will produce optimal CIS and DVS configurations for both performance-first and power-first goals, along with clear design insights indicating when each architecture is preferable.

Conclusion

This project introduces task-driven hardware optimization for CIS and DVS image sensors. By systematically sweeping design parameters and identifying optimal operating points, we bridge algorithm requirements and sensor behavior, providing practical guidance for low-power vision system design.

RAMAA's WORK:

Folder: Remaas-work (folder uploaded)

VisualComputing.ipynb (file attached)

HARSHITA's WORK:

Folder: Harshitas-work-S26-ModuCIS-modeling-main (folder uploaded)

ISH's WORK:

TODO